

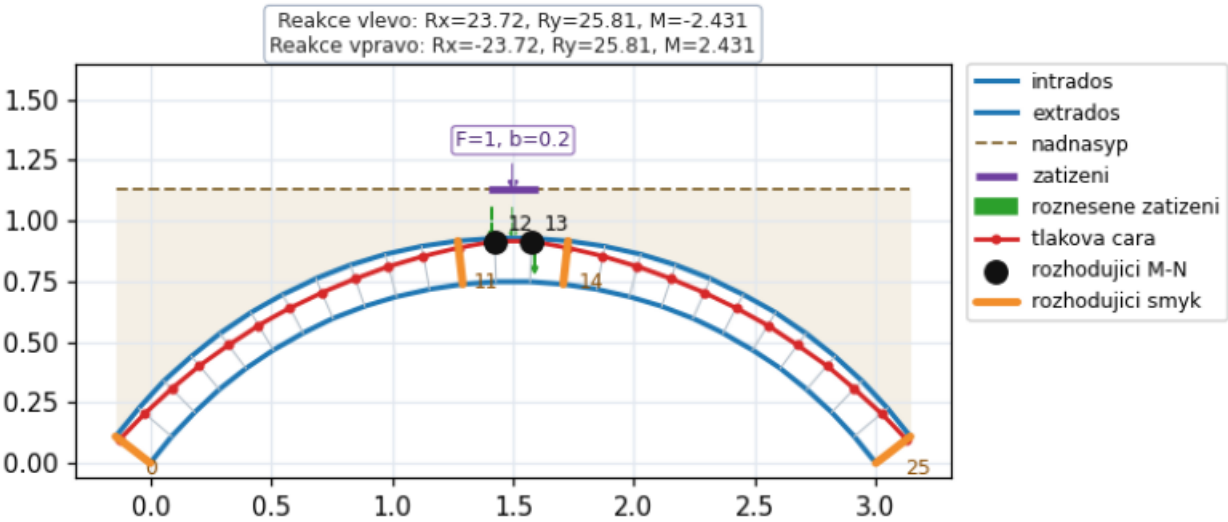
ArchLBT vypocetni protokol

1) Zadani ulohy

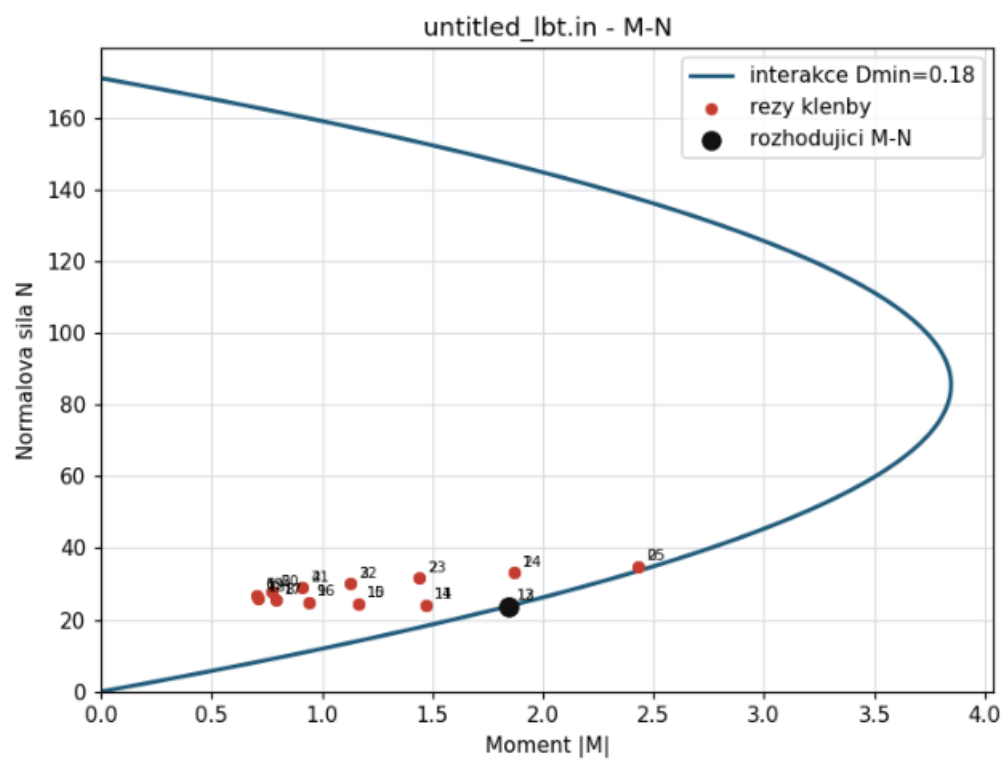
```
# ArchLBT input
# Unit convention: use one consistent force-length system.
# Geometry is in length units, typical m. Unit weights are force/volume, typical kN/m3.
# Surface point_load force F is the total force on the loaded length, typical kN.
# Self-weight and fill loads are multiplied internally by arch_width; point_load F is not.
span = 3
rise = 0.75
geom_code = 1
# extrados_mode: normal = constant normal thickness, horizontal = flat level extrados.
extrados_mode = normal
thickness = 0.18
# arch_width is the out-of-plane width b of the analysed arch strip.
arch_width = 1
masonry_unit_weight = 24
block_count = 25
point_load = 1.5 1 0.2
fill_height = 0.2
fill_unit_weight = 20
q_code = 1
fill_spread_angle_deg = 0
friction_coefficient = 0.1
compression_strength = 950
```

2) Obrazek Klenba

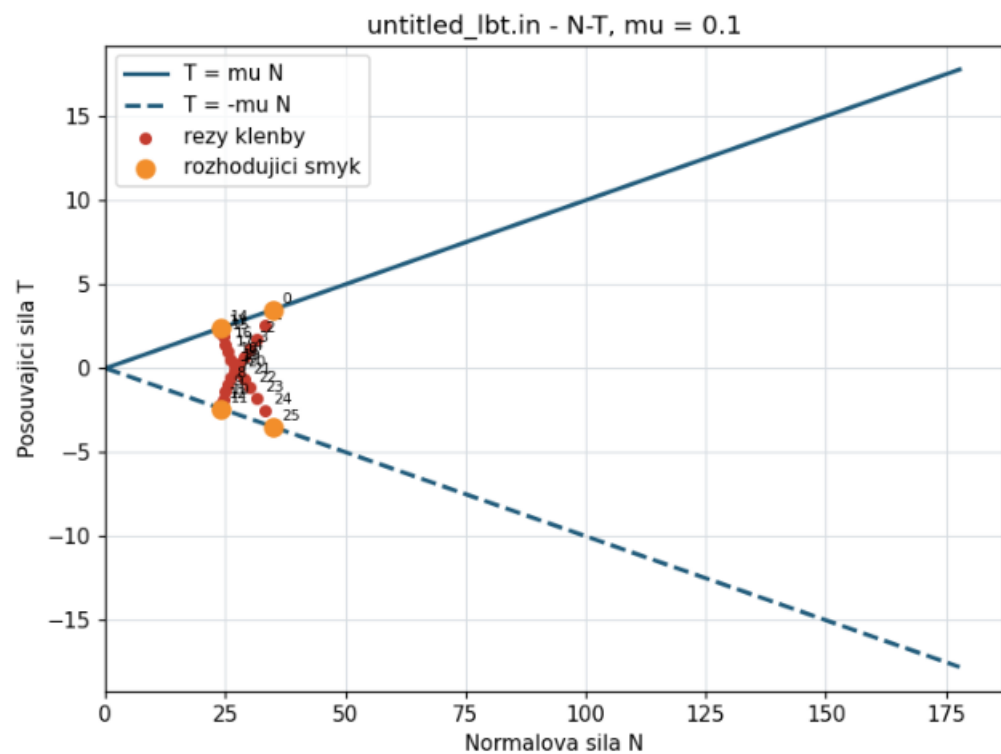
untitled_lbt.in - lambda = 6.398187



3) Graf M-N



4) Graf N-T



5) Vysledky

rez	xi	yi	xe	ye	N	M	T	e = M/N
0	-0	0	-0.144	0.108	34.882944	2.4314791	3.4882944	0.069703955
1	0.087505559	0.10807926	-0.048093908	0.22645487	33.180394	1.8668779	2.5215654	0.056264489
2	0.18278081	0.20937573	0.056327762	0.3374758	31.589832	1.4404615	1.7636005	0.045598897
3	0.28530166	0.30333222	0.16869062	0.44045212	30.147224	1.1271364	1.1653113	0.037387736
4	0.39450419	0.38943191	0.28837659	0.53481737	28.873139	0.90795273	0.67784041	0.031446277
5	0.5097877	0.46720118	0.41472732	0.6200525	27.774277	0.76993686	0.2560217	0.02772122
6	0.63051807	0.53621226	0.5470478	0.69568863	26.845712	0.70550716	-0.13878749	0.026280069
7	0.75603117	0.59608553	0.68461017	0.76130974	26.073656	0.71156071	-0.53663329	0.027290408
8	0.88563662	0.64649164	0.82665774	0.81655484	25.438534	0.78833579	-0.95773537	0.030989828
9	1.0186215	0.68715334	0.97240915	0.86112006	24.918157	0.93816155	-1.4121304	0.037649716
10	1.1542543	0.71784695	1.1210627	0.89476026	24.490763	1.1642074	-1.9001661	0.047536591
11	1.2917889	0.73840364	1.2718006	0.91729039	24.137726	1.4693381	-2.4137726	0.060873096
12	1.4304688	0.74871033	1.4237938	0.92858653	23.817541	1.843573	-2.1777275	0.077404002
13	1.5695312	0.74871033	1.5762062	0.92858653	23.817541	1.843573	2.1777275	0.077404002
14	1.7082111	0.73840364	1.7281994	0.91729039	24.137726	1.4693381	2.4137726	0.060873096
15	1.8457457	0.71784695	1.8789373	0.89476026	24.490763	1.1642074	1.9001661	0.047536591
16	1.9813785	0.68715334	2.0275908	0.86112006	24.918157	0.93816155	1.4121304	0.037649716
17	2.1143634	0.64649164	2.1733423	0.81655484	25.438534	0.78833579	0.95773537	0.030989828
18	2.2439688	0.59608553	2.3153898	0.76130974	26.073656	0.71156071	0.53663329	0.027290408
19	2.3694819	0.53621226	2.4529522	0.69568864	26.845712	0.70550716	0.13878749	0.026280069
20	2.4902123	0.46720118	2.5852727	0.6200525	27.774277	0.76993686	-0.2560217	0.02772122
21	2.6054958	0.38943191	2.7116234	0.53481738	28.873139	0.90795273	-0.67784041	0.031446277
22	2.7146983	0.30333223	2.8313094	0.44045212	30.147224	1.1271364	-1.1653113	0.037387736
23	2.8172192	0.20937573	2.9436722	0.3374758	31.589832	1.4404615	-1.7636005	0.045598897
24	2.9124944	0.10807926	3.0480939	0.22645487	33.180394	1.8668779	-2.5215654	0.056264489
25	3	4.0406878e-10	3.144	0.108	34.882944	2.4314791	-3.4882944	0.069703955

6) Report

```
ArchLBT prepared model
input = untitled_lbt.in

Geometry
span = 3
rise = 0.75
thickness D = 0.18
section depth min/max = 0.18 / 0.18
arch width b = 1
geom_CODE = 1
extrados mode = normal
blocks = 25
joints = 26

Loads
arch self-weight vertical = 15.728812
backfill vertical = 29.499758
variable vertical = 1.000000
total vertical = 46.228569

Lower-bound skeleton
state = permanent + lambda * variable
solution = statically admissible reactions Rx, Ry, M0
constraints = 0 <= N <= fc*b*D_i and abs(M) <= N*(D_i/2 - N/(2*fc*b))
shear constraint = abs(T) <= mu*N
fc = 950
b = 1
Nmax min/max = 171 / 171
mu = 0.1
status = optimal
lambda max = 6.398187
governing joint = 13
governing M-N joints = 12, 13
governing shear joints = 0, 11, 14, 25
governing compression joints = 0, 25
governing tension joints = 12, 13
Rx = 23.720402
Ry = 25.813378
M0 = -2.431479

Prepared arrays
intrados/extrados = 26 x 4
local coordinates = 26 x 4
masonry loads = 26 x 3
backfill loads = 26 x 3
variable loads = 26 x 3
```

6) Report / pokracovani

Global equilibrium check
moment origin = [0, 0]
loads sum Fx = 0.000000000e+00
loads sum Fy = -5.162675648e+01
loads sum M = -7.744013472e+01
left reaction = [2.372040163e+01, 2.581337825e+01], M = -2.431479142e+00
right reaction = [-2.372040163e+01, 2.581337824e+01], M = 2.431479142e+00
residual sum Fx = 0.000000000e+00
residual sum Fy = 0.000000000e+00
residual sum M = 0.000000000e+00

7) Generovana pole

local coordinates = 26 x 4
[[9.00000000e-02 2.49809154e+00 1.45737385e-01 8.90203409e-01]
[9.00000000e-02 2.42390793e+00 1.45737385e-01 8.16019792e-01]
[9.00000000e-02 2.34972431e+00 1.45737385e-01 7.41836175e-01]
[9.00000000e-02 2.27554069e+00 1.45737385e-01 6.67652557e-01]
[9.00000000e-02 2.20135708e+00 1.45737385e-01 5.93468940e-01]
[9.00000000e-02 2.12717346e+00 1.45737385e-01 5.19285322e-01]
[9.00000000e-02 2.05298984e+00 1.45737385e-01 4.45101705e-01]
[9.00000000e-02 1.97880622e+00 1.45737385e-01 3.70918087e-01]
[9.00000000e-02 1.90462261e+00 1.45737385e-01 2.96734470e-01]
[9.00000000e-02 1.83043899e+00 1.45737385e-01 2.22550852e-01]
[9.00000000e-02 1.75625537e+00 1.45737385e-01 1.48367235e-01]
[9.00000000e-02 1.68207175e+00 1.45737385e-01 7.41836176e-02]
[9.00000000e-02 1.60788814e+00 1.45737385e-01 1.34688756e-10]
[9.00000000e-02 1.53370452e+00 1.45737385e-01 -7.41836173e-02]
[9.00000000e-02 1.45952090e+00 1.45737385e-01 -1.48367235e-01]
[9.00000000e-02 1.38533728e+00 1.45737385e-01 -2.22550852e-01]
[9.00000000e-02 1.31115367e+00 1.45737385e-01 -2.96734470e-01]
[9.00000000e-02 1.23697005e+00 1.45737385e-01 -3.70918087e-01]
[9.00000000e-02 1.16278643e+00 1.45737385e-01 -4.45101705e-01]
[9.00000000e-02 1.08860281e+00 1.45737385e-01 -5.19285322e-01]
[9.00000000e-02 1.01441920e+00 1.45737385e-01 -5.93468939e-01]
[9.00000000e-02 9.40235579e-01 1.45737385e-01 -6.67652557e-01]
[9.00000000e-02 8.66051961e-01 1.45737385e-01 -7.41836174e-01]
[9.00000000e-02 7.91868344e-01 1.45737385e-01 -8.16019792e-01]
[9.00000000e-02 7.17684726e-01 1.45737385e-01 -8.90203409e-01]
[9.00000000e-02 6.43501109e-01 1.00000000e+00 -9.27295218e-01]]

masonry loads = 26 x 3
[[0. 0.62915246 0.02817706]
[0. 0.62915246 0.03078067]
[0. 0.62915246 0.03321497]
[0. 0.62915246 0.03546657]
[0. 0.62915246 0.03752307]
[0. 0.62915246 0.03937318]
[0. 0.62915246 0.0410067]
[0. 0.62915246 0.04241466]
[0. 0.62915246 0.0435893]
[0. 0.62915246 0.04452418]
[0. 0.62915246 0.04521414]
[0. 0.62915246 0.0456554]
[0. 0.62915246 0.04584552]
[0. 0.62915246 0.04578345]
[0. 0.62915246 0.04546955]
[0. 0.62915246 0.04490553]
[0. 0.62915246 0.0440945]
[0. 0.62915246 0.04304093]
[0. 0.62915246 0.04175059]
[0. 0.62915246 0.0402306]
[0. 0.62915246 0.03848932]
[0. 0.62915246 0.03653631]
[0. 0.62915246 0.03438234]
[0. 0.62915246 0.03203923]
[0. 0.62915246 0.02951989]
[0. 0. 0.]]

backfill loads = 26 x 3
[[0.00000000e+00 1.84671509e+00 -4.62237816e-02]
[0.00000000e+00 1.77106393e+00 -2.96265343e-02]
[0.00000000e+00 1.66529854e+00 -1.38990463e-02]
[0.00000000e+00 1.53764220e+00 1.1143965e-04]
[0.00000000e+00 1.39634000e+00 1.18512910e-02]
[0.00000000e+00 1.24944783e+00 2.10701209e-02]
[0.00000000e+00 1.10462804e+00 2.78062013e-02]
[0.00000000e+00 9.68956792e-01 3.23591958e-02]
[0.00000000e+00 8.48747066e-01 3.52460980e-02]
[0.00000000e+00 7.49391677e-01 3.71453256e-02]

7) Generovana pole / pokračovani

```
[ 0.00000000e+00 6.75229624e-01 3.88321756e-02]
[ 0.00000000e+00 6.29438900e-01 4.11095804e-02]
[ 0.00000000e+00 6.13958141e-01 4.47383270e-02]
[ 0.00000000e+00 6.29438900e-01 5.03709025e-02]
[ 0.00000000e+00 6.75229624e-01 5.84929099e-02]
[ 0.00000000e+00 7.49391676e-01 6.93755694e-02]
[ 0.00000000e+00 8.48747066e-01 8.38422022e-02]
[ 0.00000000e+00 8.68956792e-01 9.25082202e-02]
[ 0.00000000e+00 1.10462804e+00 1.17494072e-01]
[ 0.00000000e+00 1.24944783e+00 1.37016788e-01]
[ 0.00000000e+00 1.39634000e+00 1.56850491e-01]
[ 0.00000000e+00 1.53764220e+00 1.75863165e-01]
[ 0.00000000e+00 1.65629854e+00 1.92821813e-01]
[ 0.00000000e+00 1.77106393e+00 2.06464537e-01]
[ 0.00000000e+00 1.84671509e+00 2.15578313e-01]
[ 0.00000000e+00 0.00000000e+00 0.00000000e+00]
```

```
variable loads = 26 x 3
```

[illegible]